

Evaluating the efficiency of existing waste Management strategies by using GIS in Sheffield.

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ABSTRACT

This study uses Geographic Information System (GIS) technology to examine the effectiveness of current trash management practises in Sheffield. The study's objectives are to evaluate the efficacy of present waste management techniques, identify areas that could be improved, and offer suggestions for new tactics. The goals of the project include assessing waste management effectiveness, using GIS to map and analyse waste data, creating sustainable recommendations, advancing public awareness, and contributing to global waste management plans. The methodology uses a multidimensional strategy that combines field surveys, questionnaire surveys, and GIS analysis. The mapping of trash distribution made possible by GIS technology provides information on spatial patterns and disparities. Expert opinions are gathered using questionnaire surveys, indicating the perceived efficacy of waste management procedures and the use of GIS. By linking trash generation to demographic parameters, the field survey further enhances the study.

The results revealed that different parts of Sheffield have distinct distribution patterns for trash. Despite different population densities, GIS technology exposes locations with substantial garbage generation. The study emphasises the value of customised waste management plans that take into account regional conditions, economic activity, and infrastructure accessibility. Sustainable practises, community involvement, and the use of GIS for well-informed decision-making are stressed in recommendations. This study aids in improving waste management techniques in Sheffield and elsewhere. This research makes it easier to build effective, long-lasting, and community-focused waste management policies by tackling the intricacies of trash distribution and the effects of technology. It emphasises the solutions' global applicability, raising public awareness and promoting sensible waste management for a more sustainable future.

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CHAPTER 1: INTRODUCTION

Globally, waste management has become a pressing concern due to factors such as rising population and consumption, as well as environmental degradation caused by improper refuse management (Al-Dailami et al., 2022). Furthermore, the limited availability of natural resources emphasises the need for more effective waste management practises, such as waste reduction and recycling (Asiaei et al., 2021). Sheffield, a city managed by the local government and public agencies, faces the challenge of reducing waste generation and improving recycling rates (Sheffield, 2023). This research seeks to utilize Geographic Information System (GIS) technology to assess the effectiveness of current waste management practices and identify areas for improvement.

1.1 Research Objectives

The general research objectives for this study are as follows:

1. To assess the effectiveness of current waste management practices in Sheffield and identify areas for improvement.
2. To utilize GIS technology to map and analyze waste management data, providing a comprehensive understanding of the current situation.
3. To develop recommendations for improving waste management in Sheffield, considering factors such as cost, sustainability, and community involvement.

4. To contribute to the development of more efficient and sustainable waste management strategies, not only in Sheffield but also in other cities and communities worldwide.
5. To enhance public awareness and understanding of the importance of efficient waste management and encourage community involvement in the development of more sustainable practices.

1.2 Research Questions

The specific research questions that will guide this investigation include:

1. How effective are the current waste management practices in Sheffield?
2. What are the key areas for improvement in waste management strategies in Sheffield?
3. What are the spatial patterns and trends of waste generation, collection, and disposal in Sheffield?

1.3 Study Area

The study area for this research is Sheffield, a city that has implemented various waste management programs and initiatives. The waste management system in Sheffield involves waste collection and disposal services, recycling facilities, and waste treatment plants. The city aims to reduce waste and increase resource recycling. The research will focus on evaluating the waste management practices within Sheffield's jurisdiction.

1.4 Rationale

The rationale for conducting this research is based on several factors:

1. Growing population and consumption leading to increasing waste generation: The city of Sheffield faces the challenge of managing the growing amount of waste generated by its population.

2. Improper waste management causing environmental degradation: Inadequate waste management practices can lead to pollution, environmental degradation, and health hazards.
3. Limited natural resources requiring more efficient use: With finite natural resources, it is crucial to implement effective waste management strategies to maximize resource utilization.
4. Need for sustainable solutions in waste management, including waste reduction and increased recycling: Sustainable waste management practices are essential for minimizing environmental impact and promoting a circular economy.

1.5 Methodology and Methods

The research will follow a systematic methodology to achieve the research objectives. The methodology will involve the following stages:

1. **Data collection:** Data on waste production, collection, disposal, and recycling in Sheffield will be collected through surveys, site visits, and ArcGIS software. Secondary data from sources like natural earth data and the UNEP Environmental Data Explorer will supplement primary data.
2. **Data mapping using GIS technology:** GIS technology will be employed to visualize the spatial distribution of waste generation, collection, and disposal in Sheffield. This mapping will help identify spatial patterns, areas with high waste production, and areas lacking adequate waste disposal infrastructure. The ArcGIS software is one of the tools that will be used for the data mapping.
3. **Data analysis:** Geospatial analysis utilizes GIS capabilities for data analysis and visualization, enhancing waste management efficiency and identifying areas for improvement, thereby providing valuable insights for informed decision-making in solid waste management (Asefa et al., 2022).

4. **Development of recommendations:** Based on the data analysis and findings, recommendations will be formulated to enhance waste management in Sheffield. These recommendations may include the introduction of recycling facilities, composting initiatives, and educational programs to raise awareness.

1.6 Significance of the Research

This investigation is important for several reasons. First, it will contribute to the evaluation of Sheffield's extant waste management strategies, allowing for the identification of improvement opportunities. In addition, the findings can serve as a guide for other cities and communities pursuing more efficient and sustainable waste management policies. In addition, the research seeks to foster a sense of responsibility and encourage the adoption of environmentally friendly waste management practises by increasing public awareness and promoting community involvement.

CHAPTER 2: LITERATURE REVIEW

2.0 Introduction

Metropolitan areas worldwide are undergoing significant growth, leading to a considerable need for the resources necessary to support their prosperous economies. These resources encompass essential elements such as food, potable water, natural gas, petroleum, and electrical energy. The production of waste products, wastewater, and polluted air is an unavoidable outcome of the ongoing utilisation of resources in urban environments, as these areas function as thermodynamic systems (Tseng et al., 2020). The previous advancements in technology have not resulted in a decrease in the rate of resource consumption, indicating that enhancing efficiency may not be the most effective strategy for achieving sustainable living within the limitations of the planet. According to Hu et al. (2017), it is essential to prioritise the advancement of efficient systems that are in line with human requirements before undertaking initiatives to improve their environmental sustainability.

Historically, landfilling was extensively utilised as the predominant method for waste management. Nevertheless, a discernible change in viewpoint has emerged, characterised by a growing acknowledgement of the potential advantages linked to waste. The predominant method of municipal solid waste (MSW) management in various Asian countries is open disposal, primarily chosen for its cost-effectiveness. The discovery of an open dump site situated in the historical area of Rajbandh, Khulna, Bangladesh, has revealed a potential hazard to the nearby population of the contamination of soil, surface water, and groundwater. The pollution observed in this context can be attributed to the existence of contaminated leachate and gases, as elucidated in the investigations carried out by Fahmida and Rafizul in 2017, as well as Pangkaj and Rafizul in 2019. The presence of heavy metal pollutants in soil and water presents potential hazards through dermal exposure and oral ingestion. Children exhibit an increased susceptibility to the health issues in question, according to research by Pangkaj and Rafizul (2019).

Farooq et al. (2002) posited that the effective management of landfill waste is a big problem for lawmakers and planners who are trying to build smart and sustainable communities. This is mostly because of how it affects land use and environmental emissions. Existing dumps can help with the growing shortage of resources around the world by storing resources that can be used well (Peng et al., 2023). To make a complete plan for garbage mining, you need to know a lot about the amount of trash and how it works (Vollprecht et al., 2021). Systems engineering can help regulatory bodies plan, carry out, and keep an eye on waste mining projects methods (Awasthi et al. 2021; Agboola et al. 2020). Finding success indicators gives city planners the tools they need to plan waste systems well, monitor them in a systematic way, and make decisions based on good information (Bibri 2022).

Microwave pyrolysis is a new way to deal with waste, and Naveenkumar et al. (2023) asserted that it can solve complicated problems effectively while also having some side effects. Geographic Information Systems (GIS) have become more and more important in the field of waste control in recent years (Ali 2020). These groups make it possible to analyse, evaluate, and develop trash management strategies (Ali 2020; Naveenkumar et al. 2023).

2.1 Waste Management

According to Anifowose et al. (2011), waste is characterised as an object or substance that necessitates disposal due to its impaired condition, deterioration, contamination, or loss of value. The substance can exhibit harmful properties and manifest in either liquid or solid form. The aforementioned categories encompass waste materials originating from both residential and non-residential settings, such as areas inhabited by humans or animals. According to the State of Vermont Agency of Natural Resources Department of Environmental Conservation (2012), solid waste is defined as tangible and non-free-flowing materials or substances that are unwanted and originate from human activities. The term "municipal solid waste" is used to refer to waste materials that are generated from various sources, including commercial, agricultural, and industrial activities (Singh et al., 2011). Population growth and economic development have a significant impact on the generation of solid waste in metropolitan areas (The World Bank, 2019). Consequently, urban regions, small enterprises, and commercial operations commonly generate municipal solid waste.

The concept of "waste management" encompasses the comprehensive range of activities associated with the systematic collection and disposal of refuse. The aforementioned activities encompass the collection, monitoring, disposal, and recycling of waste materials. Imran et al. (2020) argue that the responsible governing bodies overseeing waste management should conduct these operations in an environmentally-friendly manner. Reducing costs is a crucial element in efficiently organising waste management operations. According to Ahmad and Kim (2020), there is a growing concern among state and local governments regarding the implementation of environmentally sustainable strategies for waste management. This is driven by the recognition of the detrimental impacts that littering imposes on the overall welfare of humans, animals, and the natural environment. An additional element of effectively designed waste management involves reducing both the quantity of waste generated and the resources allocated to waste transportation, disposal, and recycling. The field of waste management has witnessed significant advancements and improvements throughout its evolution (Imran et al., 2020).

2.2 Geographic Information System

It is possible to trace the historical presence of Geographic Information Systems (GIS) in waste management back to the early phases of GIS development. This points to the substantial use of GIS in the waste management industry. The integration of Geographic Information System (GIS) technology has led to significant breakthroughs in the organisation, implementation, and oversight of waste management activities.

Geographic information systems, or GISs, are technologies that are used extensively and which display a high level of efficiency in the process of data collecting, modification, analysis, and representation on a geographic scale. The aforementioned systems successfully combine analytical methods with geographic data, which results in a comprehensive and unified approach to the management of geographic information (Ali, 2020; Sritart& Miyazaki, 2022). There is a substantial possibility that the deployment of tools for the analysis of geographic data could make the job of researchers and practitioners in a variety of sectors easier. According to Putri (2023), these technologies can help in the organisation and analysis of geographic data, which in turn supports the formulation of informed decisions and the efficient solving of problems.

Within Geographic Information Systems (GIS) applications for the evaluation of waste management, the usage of technology is employed to optimise waste management operations, assess environmental impacts, and improve overall waste management strategies (Yu et al., 2021). When evaluating waste management practices, it is common practice to utilise Geographic Information Systems (GIS) to conduct an analysis of the spatial context of various patterns of trash generation and disposal. The use of Geographic Information Systems (GIS) makes it possible to recognise spatial patterns and trends in the production of trash and its disposal. The use of a variety of data sources is critical to the success of this endeavour. These sources include population distribution, waste generation rates, locations of landfills, and transportation networks (Khan and Samadder, 2014; Sulemana et al., 2018). The usage of this data makes it possible to optimise waste collection routes, more effectively allocate resources,

and identify regions with higher rates of waste generation that call for targeted initiatives to reduce waste (Maiurova et al., 2022).

When conducting an analysis of the effects that various waste management strategies have on the surrounding environment, the application of Geographic Information Systems (GIS) is of the utmost significance. Researchers are now able to evaluate the potential dangers and vulnerabilities associated with garbage disposal sites because of the implementation of Geographic Information Systems (GIS). Utilising Geographic Information Systems (GIS), which have the capability to superimpose several different layers of geographical information (Valenti and Toscano, 2021; Ding et al., 2022), allows for the accomplishment of the aforementioned goal. Both Abushammala et al. (2022) and Asori et al. (2022) conducted academic research in which they focused their attention on a variety of issues relating to trash management. The aforementioned concerns include the proximity of waste disposal facilities to ecosystems that are prone to harm, the potential dangers related to the contamination of groundwater, and the possibility of air pollution resulting from the operation of garbage incineration plants. According to the findings of research carried out by Sztubecka et al. (2020) and Torkayesh et al. (2021), taking into account spatial correlations can help facilitate the selection of an appropriate location for a waste facility and the implementation of mitigation measures.

In addition, Geographic Information Systems (GIS) make it possible to monitor and keep track of the procedures effectively involved in waste management. The use of Geographic Information Systems (GIS) makes it possible to visualise and analyse a variety of aspects relating to waste management. Some of these factors include the routes that garbage collection trucks take, the capacities of landfills, and recycling rates. According to Kam et al. 2021 and Mohamed et al. 2023, it has been possible to perform the integration of real-time data obtained from sensors and monitoring devices smoothly. The exploitation of these data may result in improved resource allocation, enhanced optimisation of service delivery, and streamlined procedures for waste management. The utilisation of Geographic Information System (GIS) tracking systems enables the localization of places that are lacking in waste management services through the process of identification. As a result, this makes it possible to execute targeted interventions to

improve waste management practises in underserved communities (Choi et al., 2020; Rezvani et al., 2023; Hayat, 2023) (Choi et al., 2020; Rezvani et al., 2023; Hayat, 2023).

In addition, the utilisation of Geographic Information Systems (GIS) can provide a spatial context for determining the economic viability of a variety of waste management strategies by providing a map of the area. The aforementioned citations by Paul and Bussemaker (2020) and Torkayesh et al. (2022) help make it easier for people to make educated decisions on the management of garbage. The Geographic Information System (GIS) provides users with the ability to do cost-benefit analyses as well as scenario modelling. The inclusion of data relevant to the costs involved with trash treatment and transportation, in addition to the revenues gained from recycling and energy recovery, will make it easier to achieve this goal (Kouassi, 2022). (Torkayesh et al., 2022; Meng et al., 2023) The methodology that was offered in this study enables the assessment of the financial repercussions associated with alternative waste management systems. This, in turn, empowers decision-makers to make educated choices regarding the most economically viable option.

The utilisation of Geographic Information Systems (GIS) in the evaluation of waste management has the potential to additionally contribute to the enhancement of emergency preparedness and response efforts. The integration of Geographic Information Systems (GIS) with other sources of geospatial data enables the identification of locations that are prone to natural disasters, such as floods or wildfires. As demonstrated by the research carried out by Rezvani et al. (2023), the extra data sources consist of population density maps, infrastructure placement maps, and danger map datasets. The aforementioned data can be put to use during times of crisis to devise alternate tactics, enhance emergency response routes, and maximise the utilisation of available resources (Caprioli and Bottero, 2021). The application of risk assessment models that are based on Geographic Information Systems (GIS) can assist in the identification of appropriate locations for the secure disposal of hazardous waste materials (Qian et al. 2021; Yoo and Choi 2019). This can help to address the possibility of environmental contamination during emergency scenarios.

The incorporation of Geographic Information Systems (GIS) applications in the assessment of waste management provides a valuable instrument for optimising the effectiveness of waste management protocols, evaluating the ecological consequences, and facilitating well-informed decision-making (Su et al., 2021; Palafox-Alcantar et al. 2020). Geographic Information Systems (GIS) provide researchers and practitioners with advanced functionalities for understanding waste generation patterns, assessing the environmental risks associated with waste disposal, monitoring waste management activities, evaluating the economic viability of waste management approaches, and improving disaster preparedness and response (Otero Pena et al. 2022). Since sustainable waste management practices are becoming increasingly important, the integration of Geographic Information Systems (GIS) in the evaluation of waste management demonstrates tremendous promise for optimising efficiency and reaping advantages (Akram et al. 2021).

2.3 Theoretical Framework

The term "waste management" refers to the practice of organising, planning, and effectively controlling waste. This process involves the utilisation of several different conceptual frameworks and organisational models. The hierarchy of waste management is a well-established notion that emphasises the prioritising of waste prevention, reduction, and recycling over other types of trash disposal. (European Commission, 2021) The term "waste hierarchy" refers to a structured framework that is used to prioritise waste management techniques to decrease environmental repercussions and improve the efficacy of humanitarian activities (Mussa and Suryabhagavan 2021). The framework was developed by the European Union.

2.3.1 Waste Hierarchy Framework

The waste hierarchy is an organised framework for managing trash, which is generally depicted as a sequential succession of acts, precisely "reduce, reuse, recycle, and dispose of." The term "dispose" is typically reserved for the final step in the hierarchy. According to research that was issued in 2021 by the European Commission, waste reduction and prevention are recognised as the most desirable choices within the hierarchy. Recycling has been chosen as the choice that is subsequently most preferable, while disposal has been designated as the least preferable one.

This philosophical viewpoint promotes the elimination of waste production to the greatest extent possible while simultaneously emphasizing the efficient recovery of resources (Yalcinkaya and Kirtiloglu 2021).

Utilising Geographic Information Systems (GIS), the waste chart can act as a helpful directing framework for the evaluation of how effective the current waste management systems are. The use of Geographic Information Systems (GIS) by decision-makers allows for the performance of spatial analysis and the visualisation of waste data (Abd-El and Smith 2019). This makes it possible for individuals to zero in on specific places where they may put into practice effective solutions for the prevention and reduction of waste. According to Aina and Atan (2020), waste reduction at the source can be efficiently done by the adoption of waste mapping, waste source identification, and targeted intervention. This is their argument.

In addition, the application of Geographic Information Systems (GIS) can be used to evaluate the practicability and effectiveness of recycling programmes within the context of various waste management frameworks. Geographic Information Systems (GIS) can assist in the identification of geographical areas that display a shortage of recycling facilities and also can indicate regions where more resources would be most efficiently invested (Okot et al. 2019). The incorporation of data pertaining to locations of garbage collection and recycling facilities, in addition to population density, can make the accomplishment of this goal much simpler. The research that was carried out by Asefa and colleagues (2022) offers useful information that may be used for decision-making about the allocation of resources and the optimisation of recycling efforts.

Within the context of waste management, the application of Geographic Information Systems (GIS) can be used to assess the practicability and effectiveness of recycling efforts. The use of Geographic Information Systems (GIS) allows for the determination of geographic regions that have insufficient recycling infrastructure as well as the identification of geographical areas that would be ideal for the creation of new recycling facilities (Mussa and Suryabhavagan 2021). The attentive administration of extensive records pertaining to the locations of recycling facilities, waste collection sites, and population distribution is the means by which this purpose is achieved. The research that was carried out by Asefa and colleagues (2022) provides evidence in

support of enhancing recycling activities and presents significant insights that can be used to guide decisions regarding resource allocation.

The use of Geographic Information Systems, or GIS, makes it easier to investigate different methods of waste disposal concerning the waste hierarchy. Geographic Information Systems (GIS) have the ability to expertly cartographically display garbage disposal facilities and examine their proximity to densely populated regions, important natural resources, and ecologically sensitive locations (Yalcinkaya and Kirtiloglu 2021). This capability is only one of the many applications that GIS can perform. According to Belgiu and Dragut (2016), spatial analysis plays an essential part in determining the extent to which a variety of waste disposal techniques affect the surrounding environment. It also helps in locating alternate disposal methods that are less harmful to the natural environment, such as waste-to-power plants and landfill optimisation strategies.

By integrating the waste hierarchy with Geographic Information Systems (GIS), decision-makers can visually observe and assess different waste management solutions within the context of physical space (Mussa and Suryabhavagan 2021). Utilising the capabilities of Geographic Information Systems (GIS), such as spatial querying, modelling, and assessment of the current circumstances, can help with the evaluation of the efficacy and efficiency of waste management systems. According to Aina and Atan (2020), technologies based on GIS provide support for decision-making by simulating a variety of scenarios and evaluating the potential consequences of those scenarios on waste avoidance, recycling rates, and the normal performance of waste management methods.

2.4 Empirical Studies

Fadhullah et al. (2022) conducted a study to investigate the factors influencing garbage disposal behaviours and the level of importance placed on waste management practises among residents on Malaysia's East Coast. The researchers observed the waste control practices and perceptions of families in the region. The results of the study indicated that 18.3% of households engaged in the disposal of plastic products as waste, while 74.3% of households engaged in the disposal of food debris as waste. The study additionally demonstrated that a

significant proportion of families, specifically 50.3%, engaged in waste separation practises, whereas 49.7% of families did not partake in such activities. The study identified specific areas in need of improvement and developed targeted interventions aimed at promoting sustainable waste management practices. Approximately 95.9% of the participants demonstrated awareness regarding the correlation between inadequate waste management practices and the onset of diseases such as malaria and diarrhoea. A comparable study might be undertaken in Sheffield to evaluate the various elements that influence garbage disposal practises among the local population. By gaining insights into the waste separation behaviours and disposal patterns of the local population, it becomes possible to devise specific interventions aimed at fostering sustainable waste management practises that are in line with the environmental objectives of the city.

The global COVID-19 pandemic has introduced complexities to waste management practices worldwide, exacerbating existing challenges posed by population growth and urbanisation. The global pandemic has resulted in a surge in the development and implementation of medical waste technology, thereby necessitating the incorporation of additional safety measures into waste management strategies (Chisholm et al., 2021). Chisholm et al. (2021) offered valuable insights in their narrative review on the sustainable management of scientific waste in developing African countries, providing valuable recommendations for enhancing waste disposal practices. To address the imperative of safeguarding the environment and promoting the well-being of future generations in Africa, this study examined the various aspects of sustainability in the management of medical waste in the region. The research findings present policies, potential recommendations, and solutions about sustainability and the management of medical waste, by utilising environmentally friendly technologies for efficient treatment and disposal methods. Additionally, they possess the ability to serve as intermediaries between the healthcare system, decision-makers, and stakeholders in the process of formulating health policies and implementing health programmes. The findings of Chisholm et al. (2021) regarding medical waste management in African countries offer valuable insights that can potentially inform Sheffield's waste plans in light of the dynamic issues arising from the COVID-19 pandemic. The implementation of sophisticated medical waste technologies and safety

protocols has the potential to significantly impact Sheffield's waste management strategy, thereby safeguarding both public health and the environment.

Nwosu and Chukwuemeka (2020) conducted a study to investigate the factors contributing to the low efficacy of certain management strategies. Their findings suggest that several factors play a role in this phenomenon, including population growth, the extent of the area under consideration, the level of urbanisation and industrialization, the duration of governmental tenure, an overreliance on government for waste management and financial support, as well as various environmental factors. The incorporation of the Waste Minimization approach (WMS) and the Technological Strategy (TcS) into the existing strategy has demonstrated a more effective and promising alternative, despite the prevalent reliance of waste authorities on the Traditional Solid Waste Management Strategy (TSWMS). The researchers' discoveries resulted in the suggestion that the Nigerian SWM Authorities and Agencies should consider and embrace a plan for the execution of a sustainable integrated solid waste management strategy. The research conducted by Nwosu and Chukwuemeka (2020) provides valuable insights into the various elements that can impact the effectiveness of waste management. These findings have implications for understanding the potential issues that Sheffield City may encounter in this regard. Sheffield has the opportunity to enhance its waste management effectiveness by examining various factors such as population growth, urbanisation trends, governmental policies, and reliance on traditional waste management strategies. By considering these factors, the city can explore the integration of innovative approaches such as Waste Minimization and Technological Strategies.

The plan is shown in the figure below:



Fig 1: Road Map (Nwosu and Chukwuemeka 2020)

2.5 Conceptual Framework

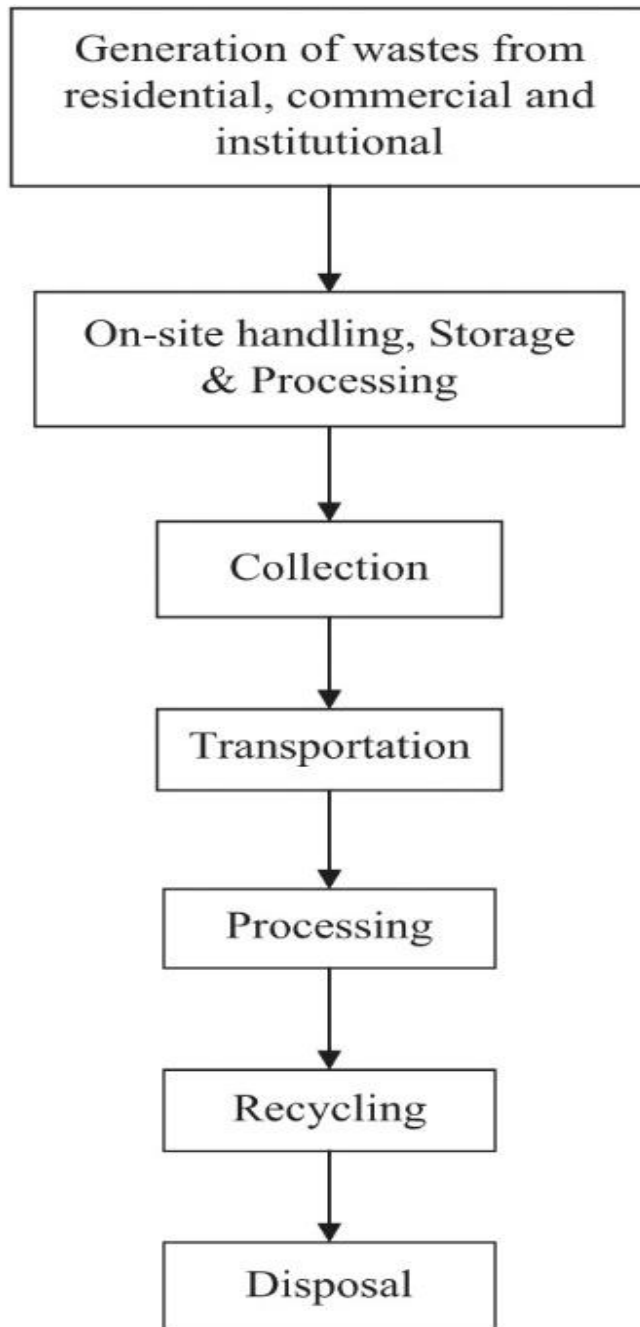


Fig 2: Conceptual framework showing the stepwise strategies for waste management. (Khan and Sukha 2014)

CHAPTER 3: METHODOLOGY

3.0 Introduction

This chapter will provide an overview of the type of research that will be carried out within this research project as well as including a methodological model. The research design is structured to comprehensively assess the efficiency of existing waste management strategies in Sheffield, utilizing ArcGIS 10.3 software for spatial data analysis. The methodology integrates spatial and attribute data acquired through field surveys, questionnaire surveys, and secondary data from literature reviews. The image (base map of Sheffield city) was georeferenced and digitized showing the location of existing waste disposal points. ArcGIS 10.3 software was used to create a database and shape files were also created and on-screen digitization was done. The study adopt field survey and survey research design for the acquisition of spatial and non-spatial data accordingly.

3.1 Data Acquisition

3.1.1 Primary Research

This research project will consist of both primary and secondary data. Primary data, also known as 'Raw Data', is data that is collected directly from a source for the purpose of research. The process of generating novel data to investigate a particular research inquiry, employing either direct techniques like interviews or indirect techniques such as observation is known as primary data (Patel and Patel, 2019). The data utilised in this study is specifically gathered for the purpose at hand and has not been previously subjected to interpretation by any external source apart from the researcher. For this research project, primary data will be made up of questionnaire survey.

a. Questionnaire Survey

According to Geuens and De Pelsmacker (2017), questionnaires are widely recognised as a prevalent method for data gathering, leading to a common association between research and the use of questionnaires among inexperienced researchers.

One of the primary benefits associated with the utilisation of questionnaires lies in its capacity to facilitate the collection of data from a considerable number of individuals situated in diverse geographical places. This enables the acquisition of demographic profiles in a streamlined and efficient manner. This tool facilitates the concise collection of qualitative fact-based and quantitative data. Questionnaires facilitate the acquisition of data regarding the frequency of opinions, attitudes, experiences, processes, behaviours, or predictions (Taherdoost, 2016).

Nevertheless, there are potential drawbacks associated with utilising questionnaires as a means of data collection (Ha, 2022), especially when it necessitates the inclusion of open-ended questions for qualitative data gathering purposes. Researchers are occasionally confronted with challenges regarding the decision to analyse a certain response or refrain from doing so. There is a potential for replies to appear inconsequential, and practical limitations such as time availability and resource requirements may also pose challenges that could lead to a decision against doing an analysis (Dalati and Marx Gómez, 2018). Nevertheless, this situation may lead to a potential sense of ethical unease due to the missed opportunity for examination. Furthermore, the presence of ambiguity among respondents might give rise to complications, especially in the context of semi-qualitative data (Dalati and Marx Gómez, 2018). Once again, the lack of clarity may force the researcher to make a decision regarding the analysis of the data.

b. Field Survey by GPS

The problem of waste is not solely attributable to the escalation in quantities, but is largely influenced by an insufficient management system (Luhar et al., 2022). The existence of open dumps poses a significant risk to both public health and the environment. The utilisation of Geographic Information Systems (GIS) is progressively gaining recognition due to their capacity to store and analyse substantial quantities of data essential for environmental management (Lü et al., 2018). A field survey was undertaken in order to get precise spatial data pertaining to the current locations of rubbish dumps in Sheffield. The utilisation of Global Positioning System (GPS) devices was employed in order to accurately document the precise coordinates, including latitude and longitude, of garbage dump sites. The primary objective of the field survey was to

construct a spatial framework that would serve as the basis for subsequent GIS analysis. The geographical coordinates of the current trash disposal site were obtained using the field measurement or field survey technique. Google Ads was utilised to acquire the coordinates. The coordinates of the solid waste dumpsites that were identified during fieldwork were imported into ArcGIS 10.3 as a text file. Subsequently, they were converted into a shape file format in order to visually represent the precise locations of these dumpsites. The points were overlaid into the street map.

3.1.2 Quantitative Research

Quantitative research encompasses several methodological techniques, including experiments, controlled observation, surveys, and statistical analysis of documents and texts (Pawar, 2020). Experiments are conducted to investigate behaviour inside a controlled context, often in a laboratory setting, with the aim of establishing causal linkages by exact manipulation of factors. Structured observation is a research method that entails the systematic observation of behaviour based on predetermined categories (Mazhar, 2021). For instance, a researcher would observe and record several types of activities taking place inside a classroom setting. Surveys can be implemented through many methodologies, including structured interviews or questionnaires (Carpinoet *al.*, 2019). These approaches involve the use of pre-set questions to gather self-reported data about individuals' attitudes, behaviours, and beliefs. The data collected is typically in a format that allows for numerical analysis. The examination of documents and texts in quantitative research, such as content analysis, prioritises the meticulous process of coding and the subsequent statistical analysis.

a. Secondary Data from Literature Review

Secondary data refers to information and data that have been previously obtained from another source for a separate research project, yet remain pertinent to my own research investigation(Harriss et al., 2019). The secondary data utilised in this research study will primarily comprise a comprehensive literature evaluation. A thorough examination of the existing literature was undertaken in order to collect secondary data pertaining to waste

management strategies, the utilisation of Geographic Information Systems (GIS) in waste management, and the identification of optimal practises in this field. The study conducted a comprehensive analysis of scholarly articles, reports, and government documents in order to provide a contextual framework, get insights into existing tactics, and uncover deficiencies in present waste management practises.

b. Alternative Data Collection Methods

Qualitative research commonly employs interviews as a primary means of gathering data, with interviews being recognized as a fundamental method of inquiry (Moser and Korstjens, 2018). Given the ubiquity of face-to-face communication in daily life, interviews have emerged as a favoured research tool due to the comfort it affords researchers (Busetto et al., 2020).

Interviews are a valuable means of data collecting due to their ability to elicit detailed responses, in contrast to questionnaires that may rely heavily on numerical data (Gill and Baillie, 2018). The research employed semi-structured interviews to facilitate the collection of data in an efficient manner. This approach involved posing direct questions to participants, enabling the researcher to guide the discussion while allowing participants to respond based on their own understanding and expertise in the subject area. . According to Dudovskiy (2022), it facilitates the opportunity to pose supplementary and pertinent inquiries in addition to the predetermined interview framework. Nevertheless, conducting interviews presents a notable challenge in terms of transcription, as it can be a laborious and time-intensive task. Maintaining the anonymity of interview participants might pose challenges, despite the utilisation of pseudonyms, which is the prescribed approach for ensuring participant anonymity. According to the study conducted by King et al. (2019), challenges pertaining to anonymity may necessitate the exclusion of certain data from the research, hence potentially influencing the overall research findings and raising ethical concerns.

3.2 Data Integration and Analysis

The research employed an extensive field survey methodology to gather precise geographical coordinates pertaining to the various waste disposal sites. These coordinates were subsequently

included into a Geographic Information System (GIS) database. The utilisation of this technology has greatly enhanced the efficiency and efficacy of visualising and conducting geographical analysis pertaining to the distribution and extension of garbage disposal facilities. The study utilised Geographic Information System (GIS) spatial analysis methods to produce maps illustrating the locations of waste disposal sites. It also aimed to detect patterns of spatial clustering and evaluate the closeness of these sites to human settlements, infrastructure facilities, and environmental features.

The goal of this study involved utilising the ArcGIS 10.3 programme, which was built by Esri, a highly regarded organisation with expertise in geospatial technology. The software in question is a highly adaptable platform that offers users a wide range of tools and functions for proficiently manipulating and analysing spatial data. This platform enables users to generate maps and effectively tackle intricate geographical obstacles with simplicity and accuracy. ArcGIS possesses a notable advantage in its capacity to facilitate the manipulation of spatial data, hence enabling users to effectively arrange, modify, and oversee spatial datasets.

ArcGIS provides customers with sophisticated analytical functionalities that enable them to derive significant insights from their spatial data. By employing spatial analysis methodologies, individuals are able to discern regularities, associations, and tendencies within their datasets, hence facilitating well-informed decision-making and aiding in the resolution of complex geographical challenges. Moreover, ArcGIS offers a comprehensive platform for the construction of maps, enabling users to utilise a diverse range of cartographic tools in order to design visually captivating and informative maps. The maps have the capability to be tailored according to specific requirements, allowing for the integration of different data layers, symbology, and labelling choices.

The data obtained from the questionnaire survey was meticulously examined in order to get attribute data pertaining to perceptions of waste management effectiveness, utilisation of Geographic Information System (GIS) technology, and recommendations for enhancement. The purpose of this analysis was to acquire a more comprehensive comprehension of these

particular elements and to identify potential areas for improvement. The data collected from the Likert scale were quantified by assigning numerical values to each response option, which ranged from 1 to 5. This was done to enable statistical analysis of the data.

The utilisation of descriptive statistics was employed to succinctly summarise and convey the primary characteristics of the data, encompassing measures of central tendency (such as the mean and median) and measures of dispersion (such as the standard deviation and range). In the field of statistics, inferential statistics are employed to derive inferences and formulate conclusions regarding the broader population, utilising the data obtained from a representative sample.

In order to evaluate the level of agreement among professionals in the waste management field, the researchers performed a statistical analysis by computing the average scores and standard deviations. The aforementioned metrics were utilised in order to ascertain the mean level of consensus and the degree of variation among the viewpoints of the experts. The researchers sought to offer a quantitative comprehension of the degree of consensus within the waste management industry through the utilisation of these statistical indicators.

3.3 Summary Table for Methodology

3.3.1 Methods used in this study

METHODS USED IN THIS STUDY		
Methods	Strengths	Weakness
Literature Review	Accesses existing knowledge and research on waste management. Provides a historical context and background information. Can identify trends, best practices, and areas of concern.	Dependent on the availability and quality of existing literature. May not capture the most current data or local context. Potential for bias in the selection of reviewed materials.

Field Survey by GPS	Provides first-hand, on-site data. Allows for direct observation of waste management infrastructure and practices. Can capture real-time information and conditions.	Resource-intensive in terms of time and personnel. Data collection may be subject to weather or accessibility limitations. Limited in scope to the specific areas surveyed.
Questionnaire Survey	Gathers perspectives and opinions from stakeholders. Allows for a broad range of participants, including residents and experts. Can provide insights into public perceptions and attitudes.	Susceptible to response bias, leading to potentially skewed data. Relies on self-reporting, which may not always be accurate. Difficulty in ensuring a representative sample.
Georeferencing and Digitization	Ensures spatial data accuracy through precise mapping. Allows for visual representation and analysis of waste disposal points. Supports spatial data integration for GIS analysis.	Requires expertise in GIS software and cartography. Errors in georeferencing or digitization can introduce inaccuracies. Initial setup and data processing can be time-consuming.
GIS Analysis (ArcGIS 10.3)	Offers powerful spatial data analysis and visualization tools. Facilitates the identification of patterns and relationships. Supports data integration	Limited by the capabilities of the chosen software version. May require specialized training and resources. Results heavily depend on the quality of input data.

	from various sources.	
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3.3.1 Methods rejected for this study

METHODS REJECTED FOR THIS STUDY	
Methods	Justification
Remote Sensing	Remote sensing techniques, such as satellite imagery, can provide valuable data on land use, vegetation cover, and even waste disposal patterns. However, it was rejected because they might not offer sufficient detail for a localized assessment at the waste disposal point level. Additionally, the cost of obtaining high-resolution satellite imagery may exceed the project's budget
In-Person Interviews	In-person interviews with key stakeholders, such as waste management officials or city planners, could provide valuable insights. However, this method was rejected due to potential logistical challenges, time constraints, and the risk of bias in responses. It could be challenging to secure interviews with busy professionals, and their responses may be influenced by their positions or perspectives.
Aerial Surveys	Aerial surveys using drones or aircraft can offer detailed spatial data. Nevertheless, it was rejected because of regulatory restrictions and potential privacy concerns associated with

	flying drones in urban areas. Additionally, aerial surveys can be costly and may not be feasible within the study's budget.
Social Media Data Mining	Social media platforms can be a source of valuable data, including public sentiment and discussions related to waste management. However, this method was rejected because it might not provide comprehensive and representative data. It could also be challenging to verify the accuracy of user-generated content on social media.

CHAPTER 4: CASE STUDY/ANALYSIS

4.1 Questionnaire Survey

4.1.1 Role and Expertise Related to Waste Management

Role and expertise related to waste management					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	GIS Specialist	19	27.1	27.1	27.1
	Environmental Consultant	38	54.3	54.3	81.4
	Waste Management Professional	13	18.6	18.6	100.0
	Total	70	100.0	100.0	

Using GIS technology to analyse spatial data, 27.1% of respondents identified themselves as GIS specialists. Experts in evaluating environmental impacts and recommending sustainable practises, environmental consultants made up 54.3% of the workforce. 18.6% of those with experience in trash collection, treatment, and disposal held jobs as waste management professionals. Collectively, these experts offer a variety of perspectives that help evaluate waste management tactics. GIS experts have the technological know-how to map waste infrastructure. Environmental consultants contribute a viewpoint on the environment, which helps to lessen ecological footprints. Waste management experts offer useful advice for practical implementation. The distribution of roles demonstrates a thorough approach to comprehending strategy effectiveness. Through this multidisciplinary collaboration, Sheffield's waste management tactics are thoroughly evaluated, enabling wise decisions for improvement.

4.1.2 Familiarity with Waste Management Practises in Sheffield

How familiar are you with waste management practices in Sheffield?					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Somewhat Familiar	11	15.7	15.7	15.7

Very Familiar	59	84.3	84.3	100.0
Total	70	100.0	100.0	

The table above revealed the familiarity level of respondents with waste management practices in Sheffield. 15.7% of the respondents said they were somewhat familiar with waste management practices in Sheffield while 84.3% of the respondents opined that they were very familiar with waste management practices in Sheffield.

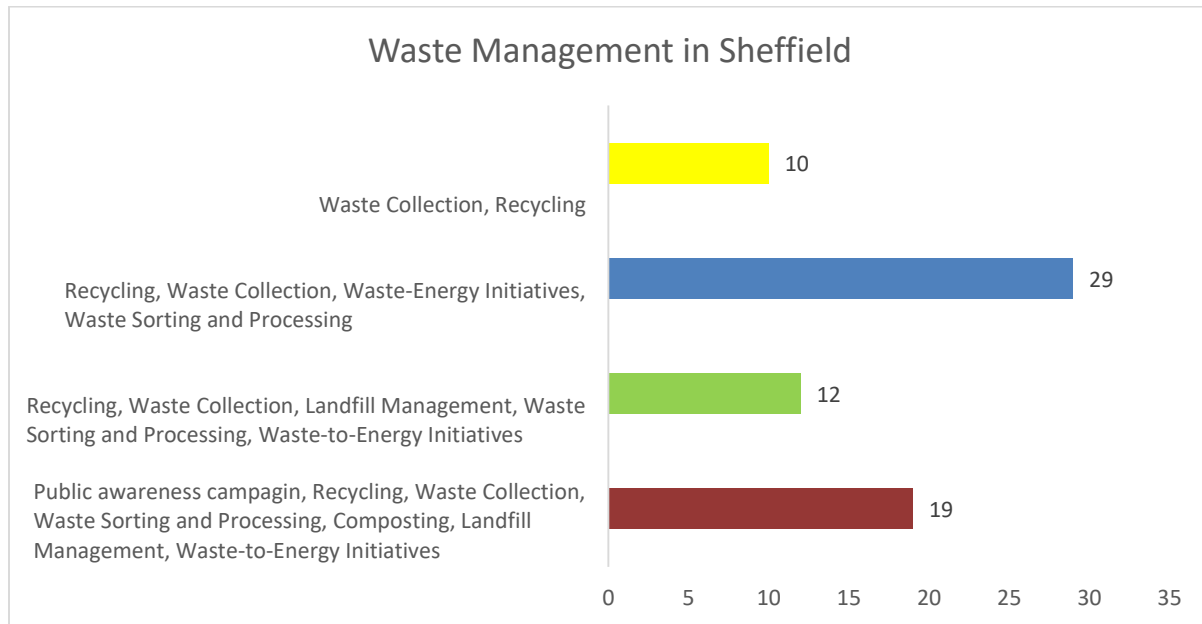
RQ1: How Effective are the Current Waste Management Practise?

4.1.3 Overall Effectiveness of Waste Management Practises in Sheffield

How would you rate the overall effectiveness of waste management practices in Sheffield?					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Ineffective	11	15.7	15.7	15.7
	Neutral	9	12.9	12.9	28.6
	Effective	24	34.3	34.3	62.9
	Strongly Effective	26	37.1	37.1	100.0
	Total	70	100.0	100.0	

The table above shows the ratings of the overall waste management practises in Sheffield. 15.7% of the respondents believed that the waste management practises in Sheffield are ineffective, 12.9% were neutral while 34.3% of the respondents opined that waste management practices in Sheffield are effective and 37.1% said waste management practices are strongly effective.

4.1.4 Which waste management aspects do you believe are currently working well in Sheffield?



The chart above shows the respondents belief in Sheffield waste management. 10 respondents believed that waste collection and recycling works well in Sheffield, 29 respondents believed that recycling, waste collection, waste-energy-initiatives, waste sorting and processing works well in Sheffield, 12 respondents believed that recycling, waste collection, landfill management, waste sorting and processing, waste-to-energy initiatives works well in Sheffield while 19 believed that public awareness campaign, recycling, waste collection, waste sorting and processing, composting, landfill management and waste-to-energy initiatives works well in Sheffield.

4.1.5 GIS Utilization and Waste Management Data

GIS technology is currently used to map waste infrastructure in Sheffield.

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Agree	24	34.3	34.3	34.3
	Strongly Agree	46	65.7	65.7	100.0

Total	70	100.0	100.0
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The table revealed the respondents agreement with the importance of GIS. 100% of the respondent agreed that GIS technology is currently used to map waste infrastructure in Sheffield.

GIS has improved the accuracy of waste collection route optimization.

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Agree	29	41.4	41.4	41.4
	Strongly Agree	41	58.6	58.6	100.0
	Total	70	100.0	100.0	

The table presented in this study provides insights into the respondents' level of agreement regarding the significance of Geographic Information Systems (GIS). The findings of the survey indicate that a unanimous consensus was reached among all participants, with a staggering 100% agreement, regarding the positive impact of Geographic Information Systems (GIS) on enhancing the precision and efficiency of waste collection route optimisation. This overwhelming level of agreement underscores the widespread recognition and acceptance of GIS as a valuable tool in the waste management industry.

GIS-based data analysis provides valuable insights for waste management decision-making.

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Agree	32	45.7	45.7	45.7
	Strongly Agree	38	54.3	54.3	100.0
	Total	70	100.0	100.0	

The table revealed the respondents agreement with the importance of GIS. 100% of the respondent agreed that GIS-based data analysis provides valuable insights for waste management decision-making.

Waste generation and disposal data are effectively visualized using GIS technology.

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Agree	32	45.7	45.7	45.7
	Strongly Agree	38	54.3	54.3	100.0
	Total	70	100.0	100.0	

The table revealed the respondents agreement with the importance of GIS. 100% of the respondent agreed that waste generation and disposal data are effectively visualized using GIS technology.

RQ2: What are the Key Areas of Improvement in Waste Management Strategies in Sheffield?

4.1.6 Recommendations for Improvement

There is room for improvement in waste management cost-effectiveness.

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Agree	25	35.7	35.7	35.7
	Strongly Agree	45	64.3	64.3	100.0
	Total	70	100.0	100.0	

According to the data presented in the table, it can be observed that a unanimous consensus was reached among all respondents, with 100% of them expressing their belief in the existence of potential for enhancement in the cost-effectiveness of waste management practises. This finding highlights the widespread recognition of the need to optimise resource allocation and operational strategies in order to achieve greater efficiency and financial viability in waste management systems.

Sustainable practices, such as waste reduction and resource recovery, should be prioritized.

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Agree	20	28.6	28.6	28.6
	Strongly Agree	50	71.4	71.4	100.0
	Total	70	100.0	100.0	

According to the data presented in the table, it is evident that a unanimous consensus was reached among all respondents, as indicated by the 100% agreement, regarding the significance of prioritising sustainable practises. These practises encompass a range of environmentally conscious actions, including but not limited to waste reduction and resource recovery. The findings of this study highlight the widespread recognition and support for sustainable practises among the surveyed individuals.

Community involvement in waste management decision-making is essential for success.

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Disagree	8	11.4	11.4	11.4
	Neutral	13	18.6	18.6	30.0
	Agree	7	10.0	10.0	40.0
	Strongly Agree	42	60.0	60.0	100.0
	Total	70	100.0	100.0	

The table above revealed that 11.4% of the respondents disagreed that community involvement in waste management decision-making is essential for success, 18.6% of the respondents were neutral, 10% agreed while 60% strongly agreed. That is, majority of the respondents recommend community involvement in waste management decision-making

4.1.7 Global Impact and Public Awareness

Insights gained from improving Sheffield's waste management can benefit other cities globally.

		Frequency	Percent	Valid Percent	Cumulative Percent
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Valid	Neutral	2	2.9	2.9	2.9
	Agree	24	34.3	34.3	37.1
	Strongly Agree	44	62.9	62.9	100.0
	Total	70	100.0	100.0	

According to the findings, a sizable majority of respondents (97.2%) either agree (34.3%) or strongly agree (62.9%) that Sheffield's waste management advances can provide lessons for other cities throughout the world. This implies a broadly held conviction in the worldwide significance and possibility for increased public awareness brought about by Sheffield's waste management improvements. Only 2.9% of respondents are still undecided about this. The strong consensus among responders confirms the belief that Sheffield's development might provide a useful example for waste management policies around the world.

Enhanced waste management practices can contribute to a more sustainable future.

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Agree	32	45.7	45.7	45.7
	Strongly Agree	38	54.3	54.3	100.0
	Total	70	100.0	100.0	

With 45.7% agreeing and 54.3% strongly agreeing, poll respondents clearly agreed on the benefits of better waste management practises. This broad consensus supports the idea that improved waste management is essential for creating a sustainable future. The findings show that there is a greater understanding of the value of ethical waste management in reducing environmental problems. The vast majority of survey participants agree that efficient waste management may make a substantial contribution to environmental sustainability, pointing to a promising trend towards increased public awareness of trash-related concerns on a global level.

Public engagement and education can foster positive changes in waste management behavior.

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Agree	37	52.9	52.9	52.9
	Strongly Agree	33	47.1	47.1	100.0
	Total	70	100.0	100.0	

According to the findings, a sizable portion of respondents are aware of the possibility of having a worldwide impact by encouraging positive changes in waste management behaviour through public involvement and education. More specifically, 52.9% concur, and 47.1% strongly concur. This may indicate a consensus among those polled that educating the public and increasing public awareness are effective ways to promote improvements in waste management. The collective agreement emphasises the idea that involved and knowledgeable citizens are essential to solving the world's problems with trash.

4.2 Field Survey (Mapping)

Urban sustainability and environmental stewardship both depend on the distribution and management of waste. Understanding the spatial patterns of garbage creation, collection, and disposal is crucial for efficient urban planning, resource allocation, and sustainable development as cities continue to expand and change. Geographic Information Systems (GIS) have become potent analytical and spatial data visualisation tools that are essential for researching garbage distribution in metropolitan settings like Sheffield.

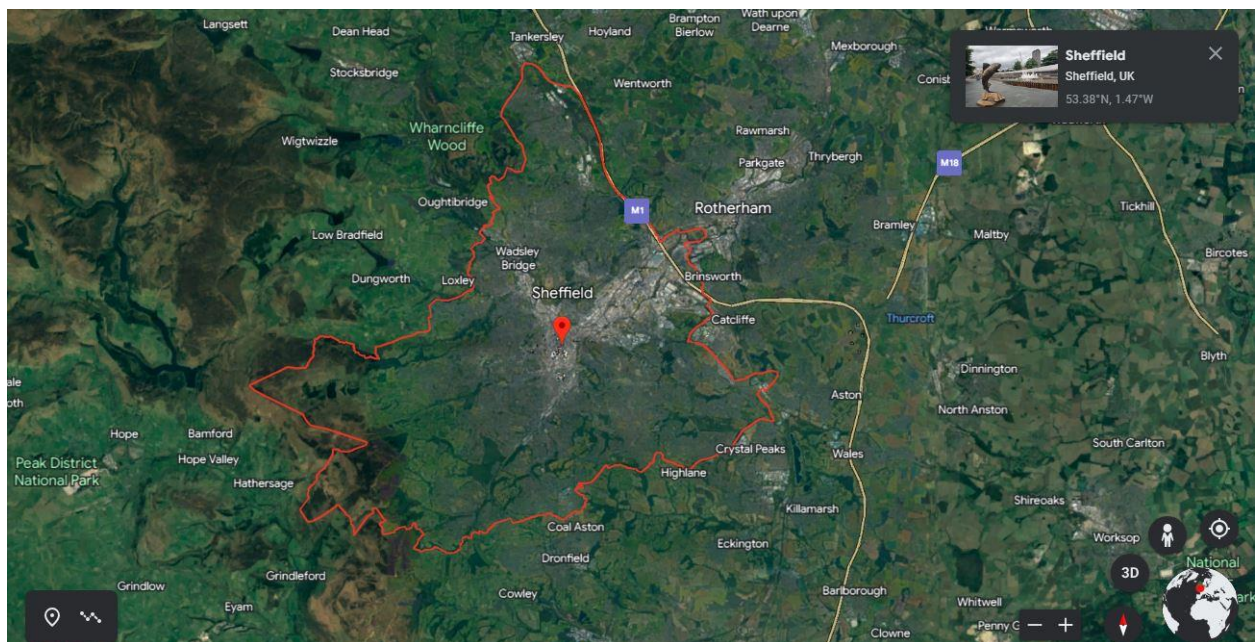
Sheffield, which is a vibrant and dynamic city in South Yorkshire, England, is distinguished by its industrial past, cultural variety, and growing metropolitan environment. Sheffield, like many urban areas, struggles to strike a balance between environmental protection and economic development. The city's trash distribution offers important insights into the complexities of these problems and potential solutions. By enabling the combination of different data sources, such as population demographics, land use, economic activities, and trash management infrastructure, GIS technology offers a novel way to comprehend garbage distribution. GIS

enables researchers, decision-makers, and city planners to unearth spatial links, spot patterns, and make well-informed decisions by leveraging the power of geospatial analysis.

GIS can determine locations with high trash generation through spatial analysis, correlate waste patterns with population densities, and evaluate the efficiency of waste collection routes. This approach can show how waste is distributed differently in various neighbourhoods, exposing potential environmental injustices. In order to facilitate effective waste management practises, GIS may also help in determining the best locations for waste facilities, recycling facilities, and disposal sites.

The goal of this part is to examine Sheffield's trash distribution patterns using GIS technology. To identify tendencies that may lead to elevated garbage buildup in particular places by analysing the relationship between waste generation and population density across diverse neighbourhoods and districts.

RQ3: What are the spatial patterns and trends of waste generation, collection, and disposal in Sheffield?



Sheffield City, UK

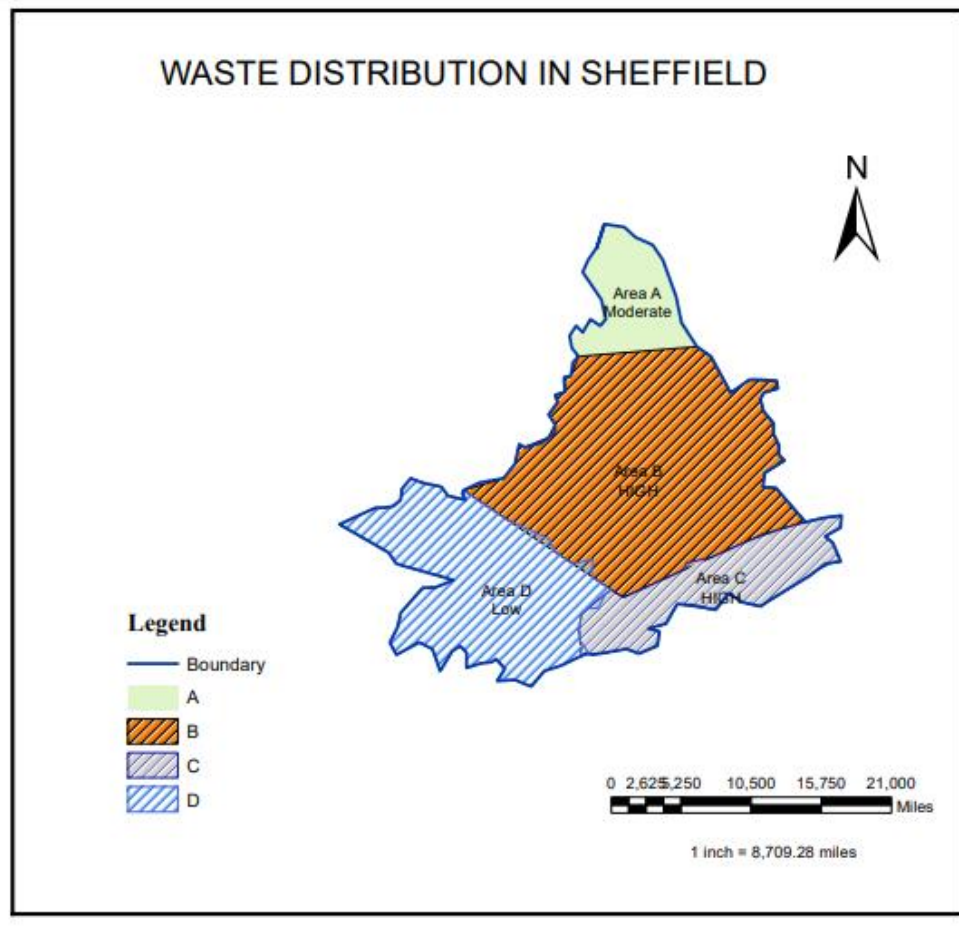
Source: Google Ads (2023)

Region	Places and population	Waste distribution
Area A	High Green= 17,489	Moderate
Area B	sheffield city centre=16,749 sheffieldhallam=93560 sheffield Hillsborough=18,605 Heeley= 21,089 Netherthorpe= 19,537	High
Area C	Shire Broke= Norton= 16,705 intake= 11,500	High
Area D	Abbeydale industrial Hamlet Stanage Pole= Wyoming brooke nature reserve=	Low (reserve areas)

Source: Author (2023)

4.2.1 Theory: Demographic Transition Theory

The Demographic Transition Theory offers an analytical framework for understanding the intricate connection between population growth and waste generation in urban settings. The literature posits that during the transition from pre-industrial to industrialised stages, there are concomitant transformations in population dynamics, urban development, and waste management practises (Nielsen, 2016; Kumar, et al., 2017).



Source: Author (2023)

Interpretation:

Area A - High Waste Generation: In Area A, encompassing various locations such as High Green, it is observed that there exists a relatively elevated distribution of waste, despite the presence of a population consisting of approximately 17,489 individuals. This observation implies that there could be various underlying factors that contribute to the increased generation of waste. These factors may include patterns of consumption, levels of commercial activity, or even the presence of inadequate waste management infrastructure. The observed moderate waste distribution within the population may suggest that although the population size is not the largest, the cumulative waste generated is still of considerable magnitude. This finding implies that even with a relatively smaller population, the waste management and disposal systems need to be adequately designed and implemented to effectively address the significant waste output.

Area B - High Population Centers: Area B, which includes significant regions such as Sheffield City Centre, Sheffield Hallam, Hillsborough, Heeley, and Netherthorpe, exhibits a substantial population size, surpassing the noteworthy threshold of 150,000 individuals. It is worth noting that waste distribution exhibits significant variation within the geographical confines of this particular region. The urban areas of Sheffield City Centre and Heeley are likely to exhibit higher levels of waste generation as a result of their bustling commercial activities and densely populated living conditions. The concentration of businesses and commercial establishments in Sheffield City Centre, coupled with the presence of numerous residential complexes in Heeley, contribute to the increased production of waste in these areas. The vibrant economic activities in Sheffield City Centre, including retail, hospitality, and entertainment sectors, generate substantial amounts of waste materials such as packaging, food waste, and discarded products. Similarly, the densely populated residential areas in Heeley, characterised by a high number of households in close proximity, contribute to the accumulation of waste from daily activities, including household waste, packaging, and other disposable items. The coexistence of multiple universities within the Sheffield Hallam area may potentially contribute to an increased generation of waste. The observed moderate waste distribution in the areas of Hillsborough and Netherthorpe may be attributed to a confluence of factors stemming from both residential and commercial activities.

Area C - High Waste Generation with Varied Population: Area C encompasses several distinct locations, including Shire Brook, Norton, and Intake. These areas, situated within a larger geographical region, exhibit unique characteristics and contribute to the overall diversity and composition of the broader area. Shire Brook, Norton, and Intake each possess their own distinct features, such as cultural heritage, socioeconomic dynamics, and urban development patterns. By examining these specific locales within the context of Area C, a more comprehensive understanding of the region. Despite the presence of a diverse population range, it is evident that there is a significant and widespread distribution of waste across these areas. This observation suggests that the generation of waste is not solely determined by

population size, but rather influenced by a multitude of factors. These factors may include industrial activities, commercial establishments, and potentially inadequate waste management practises. By acknowledging these additional influences, we can gain a more comprehensive understanding of the complexities surrounding waste generation.

Area D - Low Waste Generation: In Area D, which encompasses notable locations such as Abbeydale Industrial Hamlet, Stanage Pole, and Wyming Brooke Nature Reserve, the distribution of waste is characterised by a relatively low prevalence. The potential cause for this phenomenon can be attributed to the inherent characteristics of these regions, characterised by a lower degree of urbanisation and a reduced presence of residential and commercial infrastructure. Nature reserves and industrial hamlets often place a high emphasis on environmental conservation and the adoption of sustainable practises. As a result, these entities tend to exhibit a reduced level of waste generation. By prioritising the preservation and protection of natural resources, nature reserves and industrial hamlets actively work towards minimising their ecological footprint. Through the implementation of various strategies such as waste reduction, recycling, and the utilisation of renewable energy sources, these entities contribute to the overall goal of achieving a more sustainable and environmentally conscious society. Consequently, the commitment to environmental conservation and sustainable practises in nature reserves and industrial hamlets plays a crucial role in mitigating the negative impacts associated with waste generation, thereby fostering a more harmonious coexistence between human activities and the natural environment.

CHAPTER 5: CONCLUSION

RQ1: Current Waste Management Practice

This study has discovered significant insights in the assessment of the efficacy of Sheffield's existing waste management practises. The utilisation of Geographic Information Systems (GIS) technology, along with questionnaire surveys and field surveys, facilitated a full evaluation of

these practises. The findings of the study indicate that garbage distribution in Sheffield is characterised by non-uniformity, since different locations exhibit notable variations in the quantities of waste they generate. The aforementioned findings emphasise the importance of implementing waste management systems that are sensitive to the distinct needs and attributes of individual locations.

RQ2: Key Areas for Improvement in Waste Management Strategies

This study has obtained significant information into crucial areas for enhancement in waste management techniques in Sheffield. The questionnaire surveys conducted with waste management specialists yielded a diverse array of viewpoints and attitudes, supported by quantitative replies and subsequent statistical analysis. It is worth mentioning that there was a general agreement on the effectiveness of Geographic Information System (GIS) technology in the analysis of waste management and decision-making processes. This consensus serves to reinforce the significant role that GIS technology plays as a valuable tool in the field of urban planning and the promotion of sustainable development. Furthermore, an emphasis on sustainable practises that are ecologically conscious and actively include the community is in line with the worldwide necessity for responsible waste management.

RQ3: Spatial Patterns and Trends of Waste Generation, Collection and Disposal

In order to examine the spatial distribution and temporal dynamics of waste generation, collection, and disposal in Sheffield, the utilisation of Geographic Information Systems (GIS) technology has shown to be indispensable. By integrating several datasets that span population demographics, land usage, and trash infrastructure, this approach enabled the identification of intricate relationships and viable solutions. Significantly, this study has revealed that the generation of waste is controlled by a multitude of factors that extend beyond the number of the population. These factors encompass business activity, land utilisation, and the availability of infrastructure. The recognition that significant trash generation can occur in areas with lower human density highlights the necessity for tailored waste management strategies that consider the unique dynamics of each location.

5.1 Implications and Recommendations

The findings and inferences obtained from this research have implications that transcend the geographical boundaries of Sheffield. The comprehension of waste distribution patterns, the utilisation of GIS-based analysis, and the recognition of the significance of ecologically friendly practises are pertinent to urban centres on a global scale. The case study of Sheffield provides useful insights for decision-makers and practitioners who are faced with the task of effectively managing trash in urban areas that seek to achieve a harmonious equilibrium between environmental sustainability and economic development.

5.2 Future Research and Limitation

Future study should aim to conduct comprehensive long-term effect analyses of Sheffield's waste management practises in order to generate valuable insights into sustainability. Conducting comparative studies among metropolitan centres that encounter similar difficulties has the potential to generate cross-cultural best practises. The investigation of nascent technologies, such as improved sensors and data analytics, holds the potential to significantly improve the efficiency of waste management processes. The examination of behavioural factors related to trash creation and disposal can contribute to the advancement of responsible waste management practises. Furthermore, doing research on the interconnection between waste management and climate change mitigation presents opportunities for mitigating carbon footprints associated with trash disposal. However, it is crucial to recognise and acknowledge the limitations. The criteria encompassed in this list consist of data quality and availability, geographic breadth, time limits, resource limitations, sample representativeness, technology improvements, changing policies, and unexplored behavioural factors. In order to have a more thorough grasp of the challenges and solutions related to urban waste management, it is imperative to address these issues when pursuing future research objectives.

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Appendix

A. Data Integration and Analysis

Spatial Data Integration

The geographical data that was acquired during the comprehensive field survey, which involved the meticulous collection of coordinates pertaining to the locations of trash dump sites, were effectively integrated into a sophisticated database housed within a cutting-edge Geographic Information System (GIS). The implementation of this technology greatly enhanced the efficiency and effectiveness of visualising and conducting spatial analysis on the distribution and expansion of waste disposal sites. The research employed Geographic Information System (GIS) spatial analytic techniques to generate cartographic representations of waste disposal sites, identify spatial clustering patterns, and assess their proximity to human settlements, infrastructure facilities, and environmental attributes. By employing GIS, the study was able to effectively analyse the spatial relationships between garbage dump sites and various geographic features, allowing for a comprehensive evaluation of their distribution and potential impacts on surrounding areas. The software utilised for this purpose was ArcGIS 10.3. ArcGIS 10.3,

developed by Esri, a renowned organisation specialising in geospatial technology, represents a significant iteration of the widely employed geographic information system (GIS) software. This iteration has garnered notable recognition within the field due to its extensive functionality and capabilities. As a widely utilised tool, ArcGIS 10.3 has played a pivotal role in the analysis, visualisation, and management of geospatial data, making it an indispensable asset for professionals.

Geographic Information Systems (GIS) is a highly sophisticated technological tool that is widely utilised in various fields for the purpose of acquiring, retaining, examining, and representing geographical data and information. This advanced system enables users to effectively capture, store, manipulate, analyse, and visualise spatial data, thereby facilitating a deeper understanding of the complex relationships and patterns that exist within geographic phenomena. GIS has revolutionised the way

The ArcGIS software is a comprehensive and versatile platform that provides users with an extensive array of tools and features to effectively manipulate and analyse spatial data. This software enables users to create maps and address complex geographical challenges with ease and precision. One of the key strengths of ArcGIS lies in its ability to facilitate the manipulation of spatial data. With its diverse set of tools, users can efficiently organise, edit, and manage spatial datasets. This allows for seamless integration of various data sources, enabling users to work with a wide range of geographic information. Moreover, ArcGIS offers advanced analytical capabilities that empower users to gain valuable insights from their spatial data. Through the application of spatial analysis techniques, users can uncover patterns, relationships, and trends within their datasets. This enables informed decision-making and supports the resolution of intricate geographical issues. In addition to its data manipulation and analysis capabilities, ArcGIS provides a robust platform for map creation. Users can leverage a rich set of cartographic tools to design visually appealing and informative maps. These maps can be customised to suit specific needs, incorporating various data layers, symbology, and labelling options. Furthermore, ArcGIS offers a range of layout and publishing options, allowing users to share their maps in different formats and mediums. Overall, the ArcGIS software is a powerful and comprehensive tool that

The current study has significantly contributed to our understanding of the spatial distribution of waste disposal facilities and the potential ecological impacts associated with them. Through rigorous investigation and analysis,

this research has shed light on the intricate patterns and arrangements of garbage disposal sites, providing valuable insights into their spatial dynamics. By examining various factors such as proximity to human settlements, land use patterns, and environmental characteristics, this research has elucidated the complex interplay between waste management practises and ecological consequences. The findings of this study have enhanced our comprehension of how the spatial arrangement of garbage disposal sites can influence the surrounding ecosystems, including impacts on biodiversity, soil quality, water resources, and air pollution. Moreover, this research has also highlighted the importance of considering socio-economic factors and policy implications in understanding the spatial distribution of waste disposal facilities by examining the demographic characteristics.

Attribute Data Compilation

The data collected from the questionnaire survey was carefully analysed with the aim of gathering attribute data related to perceptions of waste management efficacy, utilisation of Geographic Information System (GIS) technology, and suggestions for improvement. This analysis was conducted in order to gain a deeper understanding of these specific aspects and to identify potential areas for enhancement. The data obtained from the Likert scale were quantified using numerical values assigned to each response option, ranging from 1 to 5, in order to facilitate statistical analysis. This quantification enables researchers to apply various statistical techniques, such as calculating means, standard deviations, and conducting inferential tests, to gain insights and draw conclusions from the data. By transforming the qualitative responses into quantitative data, researchers can effectively analyse and interpret the Likert scale data, contributing to the rigour and validity of the research findings. The data obtained from the questionnaire survey were subjected to analysis through the utilisation of both descriptive and inferential statistical techniques. These methods were employed to gain a comprehensive understanding of the collected data and to draw meaningful conclusions from it. Descriptive statistics were employed to summarise and present the main characteristics of the data, such as measures of central tendency (e.g., mean, median) and measures of dispersion (e.g., standard deviation, range). On the other hand, inferential statistics were utilised to make inferences and draw conclusions about the larger population based on the sample data.

collected. This involved the application of various statistical tests, such as t-tests or chi-square tests, to examine relationships, differences, or In order to assess the degree of consensus among experts in the waste management industry, the researchers conducted a statistical analysis by calculating the mean scores and standard deviations. These measures were employed to determine the average level of agreement and the extent of variability among the experts' opinions. By utilising these statistical measures, the researchers aimed to provide a quantitative understanding of the level of consensus within the waste management business.

B. Questionnaire

Evaluating the efficiency of existing waste Management strategies by using GIS in Sheffield.

Dear Participant,

Thank you for participating in this survey focused on waste management practices and the utilization of Geographic Information Systems (GIS) technology in Sheffield. Your valuable insights will help us assess the effectiveness of current practices, develop recommendations, and contribute to more sustainable waste management strategies. Please take a few moments to answer the following questions.

Demographic Information

1. Please indicate your role and expertise related to waste management and GIS:
 - Waste Management Professional
 - GIS Specialist
 - Environmental Consultant
2. How familiar are you with waste management practices in Sheffield?
Not Familiar | Somewhat Familiar | Very Familiar

Section 1: Current Waste Management Practices

3. How would you rate the overall effectiveness of waste management practices in Sheffield?
(Scale: 1 - Ineffective to 5 - Highly Effective)
4. Which waste management aspects do you believe are currently working well in Sheffield?
(Please select all that apply)

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4. Which waste management aspects do you believe are currently working well in Sheffield?

(Please select all that apply)

- Waste Collection
- Recycling Programs
- Waste Sorting and Processing
- Composting
- Landfill Management
- Waste-to-Energy Initiatives
- Public Awareness Campaigns

Section 2: GIS Utilization and Waste Management Data

5. GIS technology is currently used to map waste infrastructure in Sheffield.

1 (Strongly Disagree) | 2 | 3 | 4 | 5 (Strongly Agree)

6. GIS has improved the accuracy of waste collection route optimization.

1 (Strongly Disagree) | 2 | 3 | 4 | 5 (Strongly Agree)

7. GIS-based data analysis provides valuable insights for waste management decision-making.

8. Waste generation and disposal data are effectively visualized using GIS technology.

1 (Strongly Disagree) | 2 | 3 | 4 | 5 (Strongly Agree)

Section 3: Recommendations for Improvement

9. There is room for improvement in waste management cost-effectiveness.

1 (Strongly Disagree) | 2 | 3 | 4 | 5 (Strongly Agree)

10. Sustainable practices, such as waste reduction and resource recovery, should be prioritized.

1 (Strongly Disagree) | 2 | 3 | 4 | 5 (Strongly Agree)

11. Community involvement in waste management decision-making is essential for success.

1 (Strongly Disagree) | 2 | 3 | 4 | 5 (Strongly Agree)

Section 4: Global Impact and Public Awareness

12. Insights gained from improving Sheffield's waste management can benefit other cities globally.

1 (Strongly Disagree) | 2 | 3 | 4 | 5 (Strongly Agree)

13. Enhanced waste management practices can contribute to a more sustainable future.

1 (Strongly Disagree) | 2 | 3 | 4 | 5 (Strongly Agree)

14. Public engagement and education can foster positive changes in waste management behaviour.

1 (Strongly Disagree) | 2 | 3 | 4 | 5 (Strongly Agree)

